[Student title]

Formatting template – not a completed submission

Muhammad Jon Raza
[Student ID and institution]

Abstract

[Student writing: Write your abstract in your own words.]

Repository: `https://github.com/0x0shephard/atml-pa2`

1 Introduction and research questions

[Student writing: Write your introduction and research questions.]

Template only. The course requires student-written report language and analysis.

Preprint.

2 Experimental setup and methods

[Student writing: Describe the fixed datasets, policies, checkpoints, objectives, budgets, decoding, and evaluation procedures.]

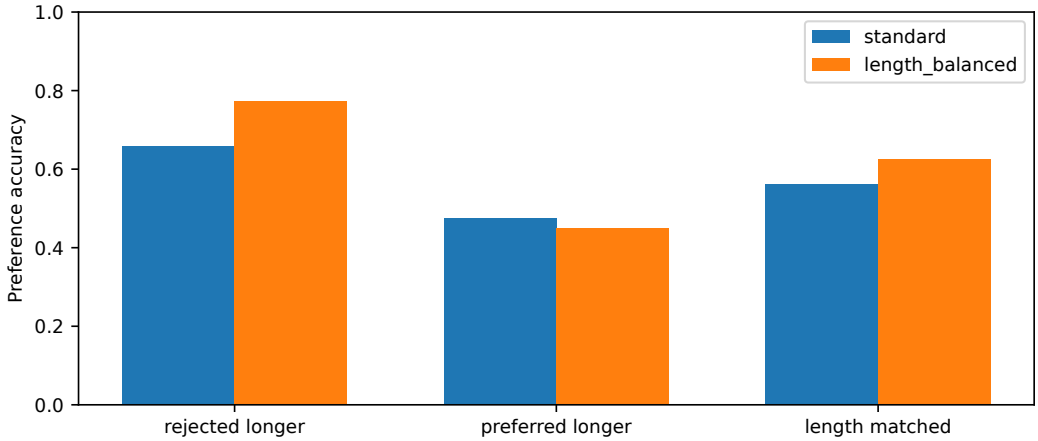
Task	Train seconds	Peak GiB/device	Epochs/updates
task1_dpo	1295	6.657	1
task2_ppo	320.1	7.408	20
task3_grpo	404.8	6.497	20

[Student writing: Write the resource-table caption and describe retained DPO sample counts, prompt-length filtering, and actual versus allocated token budgets.]

3 Task 1: DPO

Condition	Loss	Accuracy	Reward	KL	Tokens
standard	0.6741	0.6414	0.9642	-7.46e-06	168.5
beta_0.03	0.6909	0.6448	0.9254	1.855e-05	168
beta_0.10	0.6862	0.6414	0.9485	1.561e-05	167.3
beta_0.30	0.6756	0.6483	0.9485	2.742e-05	168.3

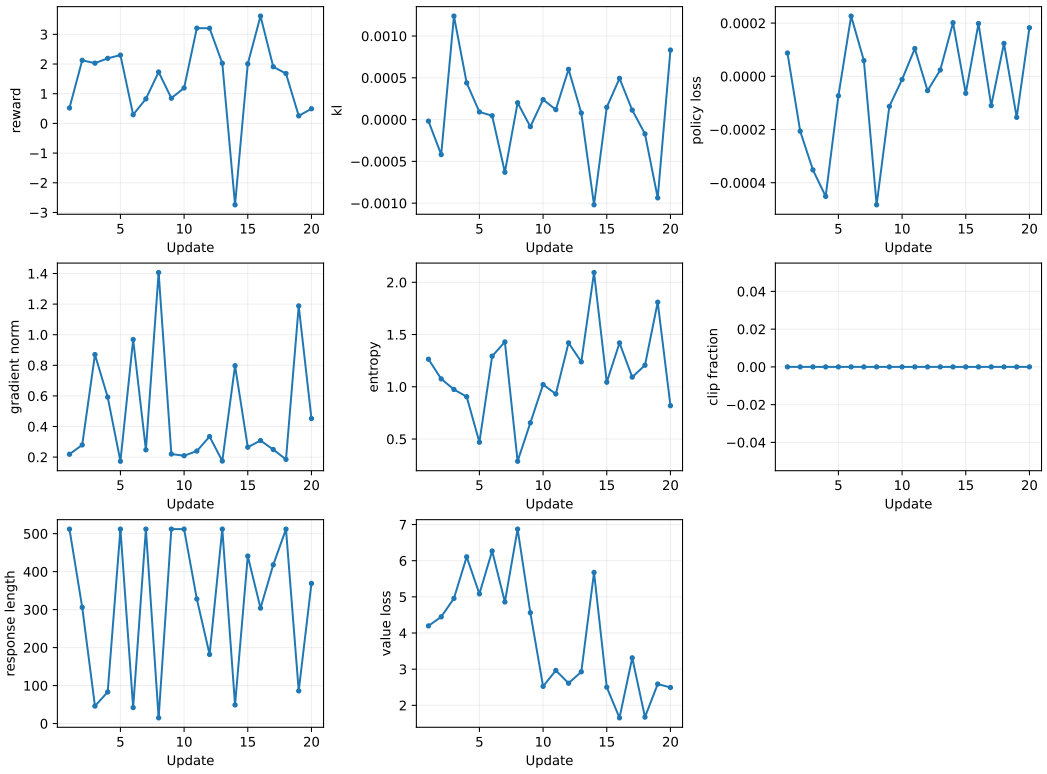
[Student writing: Write the table caption and your analysis of the standard run and beta forks.]



[Student writing: Write your own caption for the length-stratified comparison.]

[Student writing: Discuss length and word-limit evidence. Select and interpret qualitative examples from saved responses yourself.]

4 Task 2: PPO

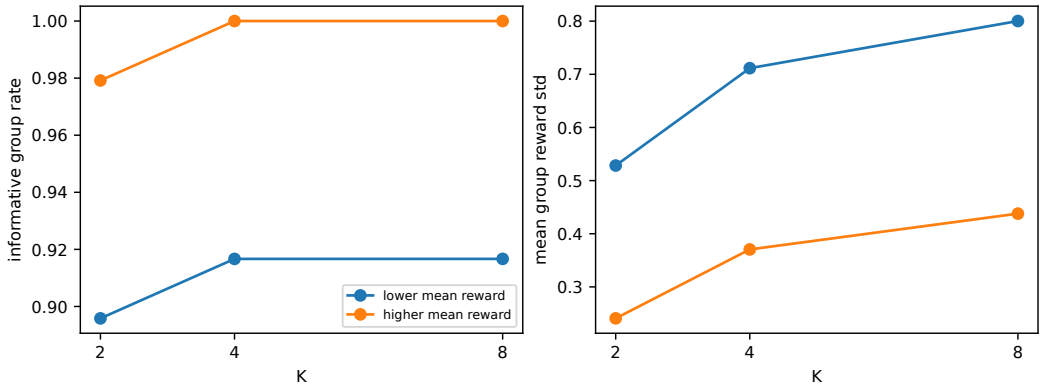


[Student writing: Write your own caption for the PPO continuation trajectories.]

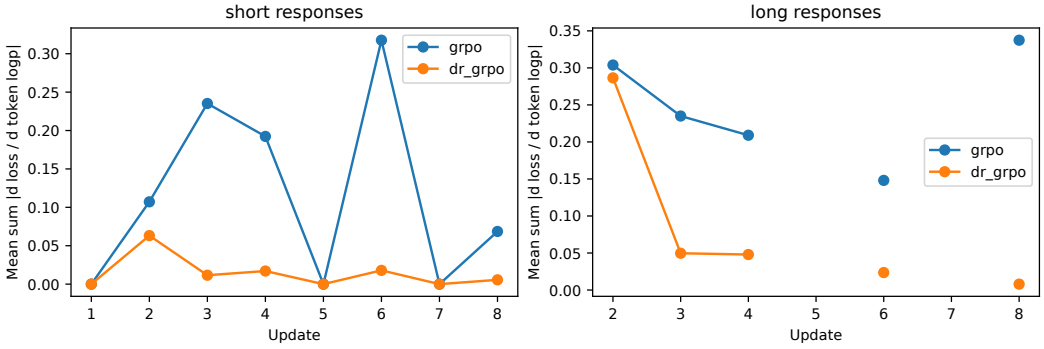
Epsilon	Surrogate loss	Affected fraction
0.05	0.005276	0.1077
0.2	0.003847	0.008509
0.5	0.003172	0.005559

[Student writing: Write the cache-table caption. Use the clipping/KL comparison tables and stability statistics for your analysis; choose brief qualitative examples yourself.]

5 Task 3: GRPO



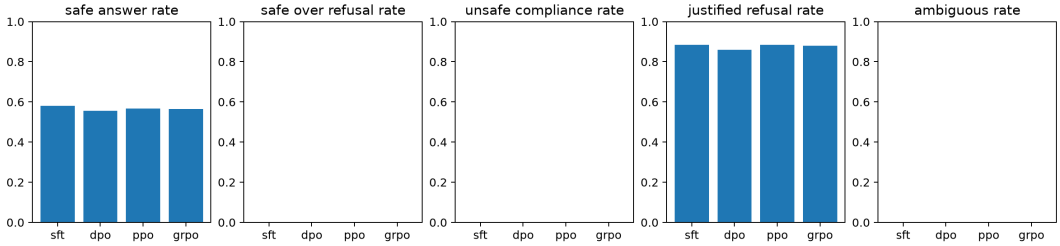
[Student writing: Write your own caption for the equal-generation group-size study.]



[Student writing: Write your own caption for the length-conditioned score-gradient comparison.]

[Student writing: Write your interpretation of normalization, held-out behavior, and qualitative examples. Use the saved normalization table and resource measurements.]

6 Task 4: Safety calibration



[Student writing: Write your own caption for the automated safety rates.]

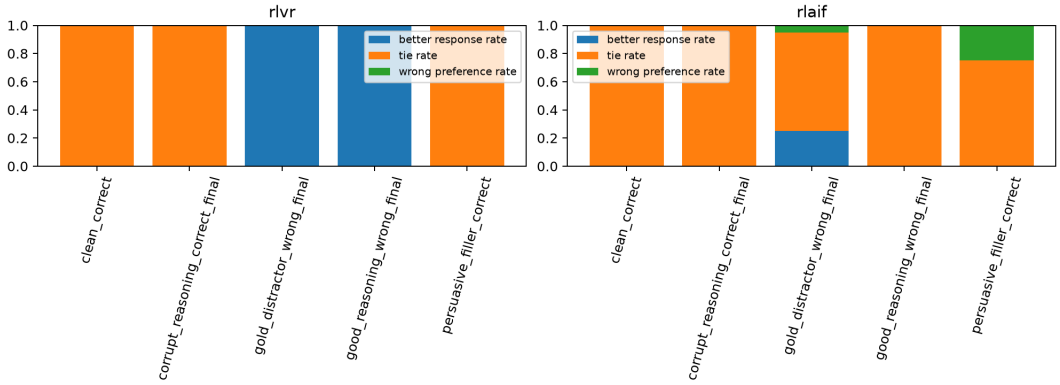
[Student writing: Complete the blind audit first. Insert your manual agreement and confusion matrix after regeneration; then discuss safety calibration and judge limitations.]

[Student writing: Select category-level evidence and brief examples from the saved records.]

7 Task 5: Feedback sources

Dataset	Policy	Accuracy	Format	Win rate	Tokens
gsm	sft	0.27	0.3467	0.5	276.4
gsm	rlvr	0.3	0.3733	0.5017	275.5
gsm	rlaif	0.2933	0.37	0.5	274.9
transfer	sft	0.47	0.6	0.5	163.9
transfer	rlvr	0.44	0.56	0.515	163.3
transfer	rlaif	0.44	0.57	0.515	163.5

[Student writing: Write the comparison-table caption and your in-domain versus transfer analysis.]



[Student writing: Write your own caption for the controlled perturbation diagnostics.]

[Student writing: Discuss sensitivity statistics and qualitative disagreement examples in your own words.]

8 Cross-task discussion

[Student writing: Write your synthesis using the manual's Task 6 research questions.]

9 Conclusion

[Student writing: Write your conclusion, limitations, and scope of the evidence.]

References

[Student writing: Insert the sources you actually read and cited. References do not count toward the course's eight-page main-content limit.]